

Part I. System Engineering

Courseware 01 - Introduction to Systems Engineering (SE)

1.1 What is a system? What is an engineered system? Cite some examples.

- A system is a set of interacting components functioning together toward same objective(s).
- A system has:
 - components,
 - attributes
 - relationships,
 - interfaces,
 - a boundary,
 - an environment.
- An engineered system is a technical or socio-technical system designed to do a defined job.
- Examples of engineered system:
 - an aircraft,
 - a control station,
 - a hammer.

Sources: Courseware 01 p.5-8, 33; Tutorial Solution p.11.

1.2 What is systems engineering?

- System engineering is a top-down, interdisciplinary approach that turning a need into a successful system.

Sources: Courseware 01 p.8, 16, 22-25, 29-31, 33.

1.3 What are user needs? What are requirements?

- User needs are function system must perform to meet user capability.
- Requirements are essential conditions and properties that the system and its elements must meet.

Sources: Courseware 01 p.22-25, 33; Courseware 06 p.16-22; 2020; 2024.

1.4 What is verification and validation?

- Verification: is the system right? $\Rightarrow$ are the requirements met?
- Validation: is it the right system? $\Rightarrow$ is the system fit for purpose?

Sources: Courseware 01 p.24-25, 29-31; Courseware 05 p.13; Courseware 07 p.21-24; Tutorial Solution p.9; 2025; 2024.

Courseware 02 / 03 / 04 - Introduction to Unmanned Aerial System (UAS)

2.1 What are the elements of a UAS? Which element performs the main function?

- Main UAS elements:
 - UAV,
 - payload,
 - control station(s),
 - support, communication, transport, ... subsystems
 - (if required) aircraft launch and recovery subsystems.
- In many UAS problems, the payload is the reason the system exists, while the rest of the UAS supports that payload mission.

Sources: Courseware 02 p.6, 18; Courseware 03 p.2; Courseware 04 p.2; 2025; 2024.

2.2 What are the main advantages of unmanned vs. manned systems?

- No onboard crew risk.
- D₃: dull, dirty, dangerous missions.
- Can stay airborne for long periods.
- Can access remote, hostile, or confined areas more easily.
- Can reduce some certification and production burden in lower-risk categories.
- Can be smaller and more mission-focused than manned aircraft.
- Can support distributed or autonomous operation.
- Can be cheaper and easier to deploy (in some missions).

Sources: Courseware 02 p.8; Courseware 04 p.30.

2.3 List typical UAS military and civilian mission needs.

- Military mission needs:
 - intelligence, surveillance, and reconnaissance,
 - weapons delivery,
 - guided missile support,
 - cargo transport,
 - search and rescue,
 - pilot augmentation,
 - location and destruction of land mines.
- Civilian mission needs:

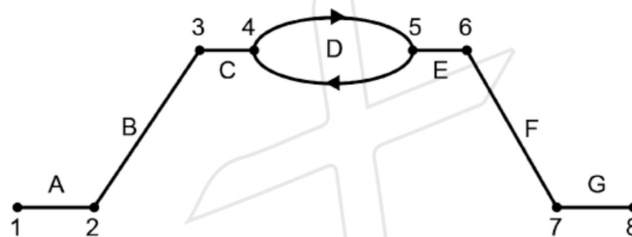
- recreational use,
- traffic management,
- point-to-point delivery,
- disaster management,
- surveillance,
- wildlife monitoring,
- aerial surveying, including LIDAR,
- precision agriculture and farming,
- construction and infrastructure inspection,
- photography, film, and television.

Sources: Courseware 02 p.23; 2025.

2.4 Sketch a typical reconnaissance mission profile of a UAS.

Stages of mission:

- Take-off run;
- Climb;
- Cruise out;
- Monitoring;
- Cruise back;
- Approach;
- Landing.



Transition points:

- Engine ignition;
- Take-off;
- Operational altitude point;
- Start monitoring;
- End monitoring;
- Start descent;
- Landing;
- Stop engine.

Diagram reference: Courseware 02 p.24. Related mission-profile solution also appears in 2024.

Sources: Courseware 02 p.24; 2024.

2.5 What is the control station element in the UAS? Name its main functions.

- The control station is the human-machine interface (HMI) of the UAS.
- Main functions:
 - present UAV status and interface with the flight control system,
 - recognize, monitor, and control the payload,
 - provide ground communication elements between control station and UAV,
 - monitor position and flight path using displays and data processing,
 - interact with external systems e.g. weather, GNSS, higher authority tasking, and reporting.
 - operating center for the off-board human role.

Sources: Courseware 02 p.26; 2024.

2.6 What form of data transmission is used in UAS communication systems?

- radio $\Rightarrow$ main practical solution, it may be direct, via satellite or via radio relay.
- laser $\Rightarrow$ much more limited by atmospheric effects and practical constraints.
- fibre-optics $\Rightarrow$ for specific task require low altitude, high data rate, and high security.

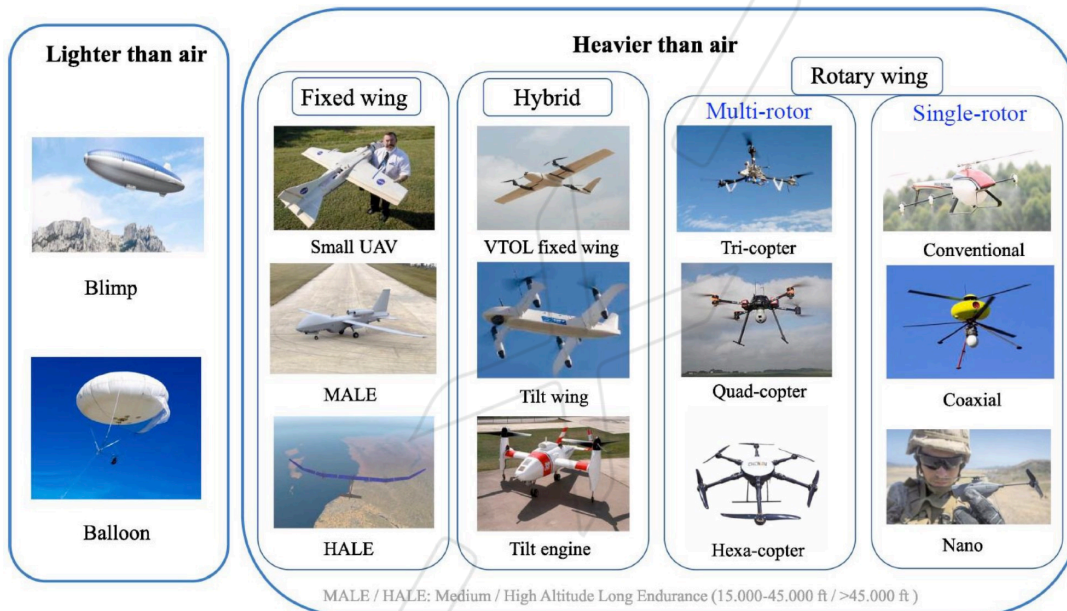
Sources: Courseware 02 p.29; 2024.

2.7 What is the main function of the UAV in the UAS?

- The UAV carries the mission payload and the subsystems required to operate.

Sources: Courseware 03 p.2; 2025; 2024.

2.8 Name all UAV configurations.



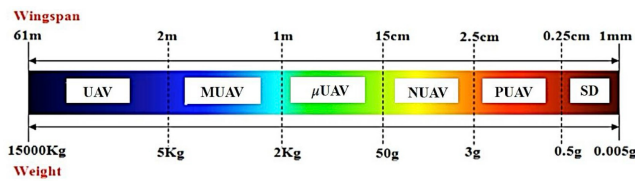
Sources: Courseware 03 p.5-9.

2.9 What is the common type of mission performed by HALE and MALE, and TUAV UAS?

- high altitude long endurance (HALE) and medium altitude long endurance (MALE):
 - long-endurance intelligence, surveillance, and reconnaissance,
 - carrying a sophisticated payload over long distances for long times.
- medium range or tactical UAV (TUAV):
 - tactical surveillance and local reconnaissance,
 - low-altitude, rapid-response missions.
 - close-range monitoring tasks.

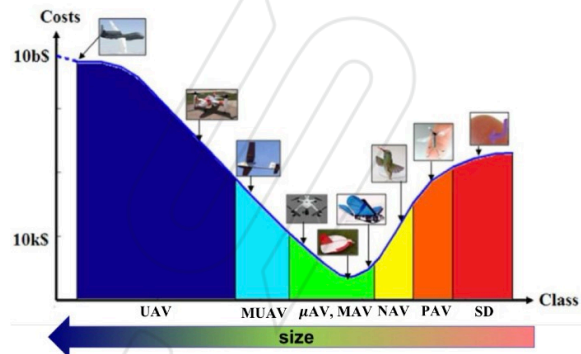
Sources: Courseware 02 p.23; Courseware 03 p.13-19; 2023.

2.10 How does cost vary with size for vehicles smaller than mUAVs, and for bigger ones?



Legend: mini (MUAV), micro (μ UAV), nano (NAV), pico (PAV), smart dust (SD)

- Why very small vehicles become expensive:
 - difficult miniaturisation,
 - difficult manufacturing,
 - fragile components,
 - advanced sensors / integration.
- Why bigger vehicles become expensive:
 - more structure,
 - more complex propulsion and avionics,
 - more demanding integration,
 - greater certification and support burden.



Sources: Courseware 03 p.12, 21; Courseware 04 p.30.

2.11 What are the typical endurance, range and altitude requirements for HALE and MALE UAVs?

- HALE: over 15000 m altitude, more than 3000 km distance, 24-48 hours endurance.
- MALE: 5000-15000 m altitude, more than 3000 km distance, 24-48 hours endurance.

Sources: Courseware 03 p.13-14, 23.

2.12 What are the main design characteristics of a mUAV? What is their typical all-up-mass?

- Main design characteristics of a mini-UAV (mUAV):
 - back-packed $\Rightarrow$ portable,
 - can be assembled and deployed by 1-2 people $\Rightarrow$ easy to deploy,
 - can be hand-launched (sometimes use catapult or rail) $\Rightarrow$ easy to launch and recover,
 - usually controlled via a laptop,
 - usually battery powered,
- Attractive for commercial and tactical short-range use.
- Typical all-up-mass: < 20 kg
 - often < 10 kg $\Rightarrow$ can fly under model aircraft rules.

Sources: Courseware 03 p.10, 13, 20-21; Courseware 04 p.31.

2.13 What are common missions of mUAVs and nUAVs?

- urban,
- indoor,
- flying within buildings,
- flying around natural structures,
- short-range surveillance,
- monitoring,
- quiet or discreet observation,
- swarm-based use,
- ultra-short-range local sensing.

Sources: Courseware 03 p.10, 20-21.

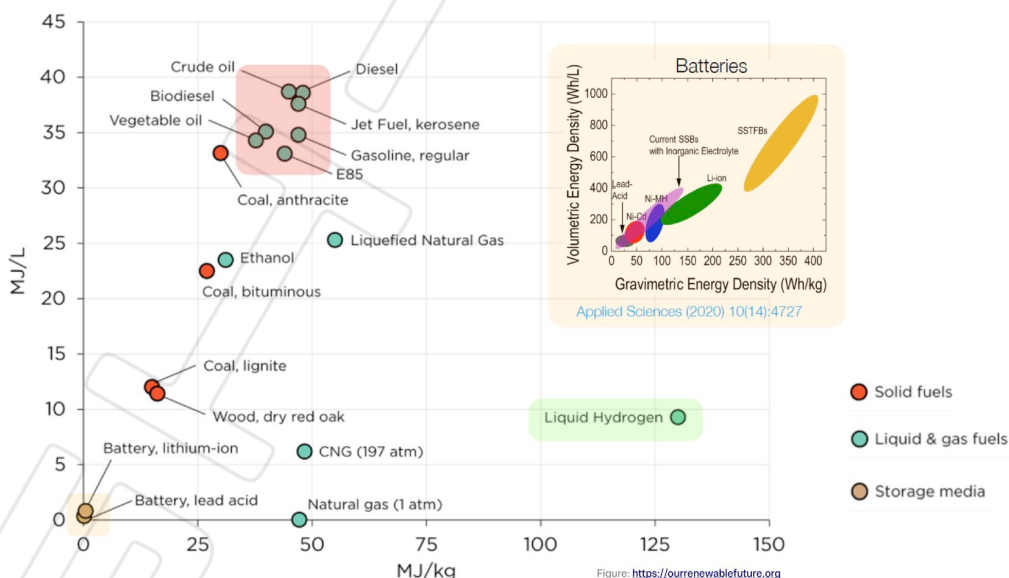
2.14 Name energies and mechanical energy converters in UAV propulsion systems.

- biochemical: fuel $\Rightarrow$ internal combustion (piston or turbine) engine,
- electrical: batteries $\Rightarrow$ electric motor,
- photovoltaic: solar energy $\Rightarrow$ electric motor.

Sources: Courseware 03 p.27; 2024.

2.15 Why are liquid fuels preferred to electrical batteries for high endurance and payload?

- Liquid fuels have much higher energy density than batteries.



Sources: Courseware 03 p.27; 2023; 2022.

2.16 What are guidance, navigation, and control (GNC) systems?

- Guidance $\Rightarrow$ motion planning: path planner, path manager.
- Navigation $\Rightarrow$ sensing and state estimation.
- Control $\Rightarrow$ commands execution and stabilisation: path following, autopilot.

Sources: Courseware 03 p.33, 36-39; 2023.

2.17 What are payload design factors?

- required platform range, endurance, and altitude,
- mass and drag constraints,
- installation constraints,
- field-of-view constraints,
- mission-dependent requirements...

Sources: Courseware 03 p.41; 2023.

2.18 When are launch and recovery systems used?

- When the UAV is not VTOL and a runway is not available.
- Launch methods:
 - hand,
 - rail,
 - catapult,
 - launch from another vehicle,
 - VTOL in hybrid systems.
- Recovery methods:
 - belly landing,
 - recovery net,
 - parachute, sometimes with airbags,
 - vertical landing in hybrid systems.

Sources: Courseware 03 p.19, 42; 2023; 2025.

2.19 What is stakeholders? Name stakeholders in the integration of UAS in the airspace system.

- Stakeholders are the components of the system that affect system or being affected by the system, which including:

- manufacturers,
- commercial vendors,
- industry trade associations,
- technical standards organizations,
- academic institutions,
- research and development centres,
- governmental agencies and other regulators,
- general public.

Sources: Courseware 04 p.6; 2016.

2.20 What are the altitude limitations for UAVs of the open and specific categories? What regulations apply to UAVs of the certified category?

- Open category:
 - explicitly limited to 400 ft (120 m) from the surface,
 - generally visual line of sight (VLOS) under basic predefined limitations.
- Specific category:
 - not one single fixed open-style altitude rule,
 - operation depends on CAA operational authorisation and a risk-based approval process.
- Certified category:
 - high-risk operations $\Rightarrow$ manned-aviation-like regulatory regime,
 - certification of the aircraft, UAS operator; licensing of the remote pilot.

Sources: Courseware 04 p.20-23.

2.21 Name human factors challenges for the design and development of UAVs.

- reduced sensory cues $\Rightarrow$ no out-the-window view $\Rightarrow$ hard to monitoring the aircraft state.
- low-quality, high-latency or interrupted radio-link control and communication.
- large or complex control station.
- transfer of control during operations.
- unconventional UAV characteristics.
- controls and displays from different COTS providers causing integration issues.

Sources: Courseware 04 p.24-25, 31.

Courseware 05 - SE for Aircraft Systems and the Aircraft Lifecycle

3.1 Stages of the lifecycle.

- CD²BTO(R²): Concept, Definition, Design, Build, Test, Operate, Retire or refurbish.
- Acquisition: Concept, Definition, Design, Build, Test ⇒ Utilisation: Operate.
- Soft: Concept, Definition ⇒ Hard: Design, Build, Test, Operate.

Sources: Courseware 05 p.9, 13, 35-37, 41; 2023.

3.2 At what stage should SE invest more effort to reduce costs?

- Mainly at the earlier (soft) stages: Concept and Definition.

Sources: Courseware 05 p.15-19; Courseware 01 p.17, 30-31; 2020.

3.3 What stage of the acquisition phase requires the largest number of engineering skills?

- The definition and early design phase, where the technical activities contribute the most.

Sources: Courseware 05 p.20-24; 2023.

3.4 What is the aim of each phase?

- Concept: define the problem, stakeholders needs, system requirements; generate candidate concepts; check role, performance, risk and manufacturable feasibility.
- Definition: turn the reviewed concept into a preliminary design; define architecture, interfaces, key performance; evaluate supplier base and technology choices; identify risks and feasibility.
- Design: refine the preliminary design into detailed subsystem and component designs that can be manufactured, integrated, tested and certified; select suppliers of bought-in equipment and components.
- Build: manufacture and assemble the system from design data, manage production queries and configuration, and perform build and acceptance testing.
- Test: prove the aircraft and its components are fit for purpose through ground/flight testing, modification, analysis and certification evidence.
- Operate: support daily operation, maintenance and logistics; collect operational data and analyse/report defects or faults.
- Retire: decide whether to store, refurbish or destroy the aircraft/components, and plan safe removal, storage, disposal and record keeping.
- Refurbish: change the aircraft role because of obsolescence or customer wishes, then return the type record to concept for a new lifecycle.

Sources: Courseware 05 p.21-38.

3.5 What is the output of each phase?

- Concept: requirements, reports, drawings, brochures, a number of potential solutions.
- Definition: feasibility reports, performance estimates (operational performance model), individual-system behaviour models, 3D/experimental models and sometimes prototypes.
- Design: detailed subsystem/component designs, engineering drawings/data, software, engineering/prototype models, supplier/production plan, and test/certification procedures.
- Build: complete product, i.e. the manufactured and assembled aircraft/system.
- Test: completed test programme, design analysis and documentation, certification evidence and database.
- Operate: customer and aircraft operations support, including defect reporting, regression-test needs and in-service support data.
- Retire: retirement plan to assist the customer, with safe removal, storage or destruction and archived design/qualification records.
- Refurbish: requirements for the next concept phase and a new lifecycle for the refurbished aircraft.

Sources: Courseware 05 p.21-38; 2023.

3.6 What stage or phase leads to certification?

- The test phase: conclusion of the test programme, together with the associated design analysis, documentation and database.

Sources: Courseware 05 p.29-31; Courseware 07 p.19, 23; 2020.

3.7 At what stage do you involve suppliers in the lifecycle?

- Suppliers should be involved from the definition stage onward.
- For complex integrated systems, early supplier involvement is important because architecture, interfaces, certification, and production planning all depend on it.

Sources: Courseware 05 p.16, 24; 2023.

- The diagonal contains the system elements or functions.
- Off-diagonal entries show directional interfaces or interactions.

Sources: Courseware 06 p.39-45; Courseware 07 p.7-9; Tutorial Solution p.3.

4.3 How do you elicit the requirements?

- Elicit via requirement analysis from stakeholder enquiries, regulations, ...

Sources: Courseware 06 p.16-22; 2021; 2024.

4.4 How do you describe or clarify the requirements?

- SMART: Specific, Measurable, Achievable, Relevant, and Timely.

Sources: Courseware 06 p.20-21.

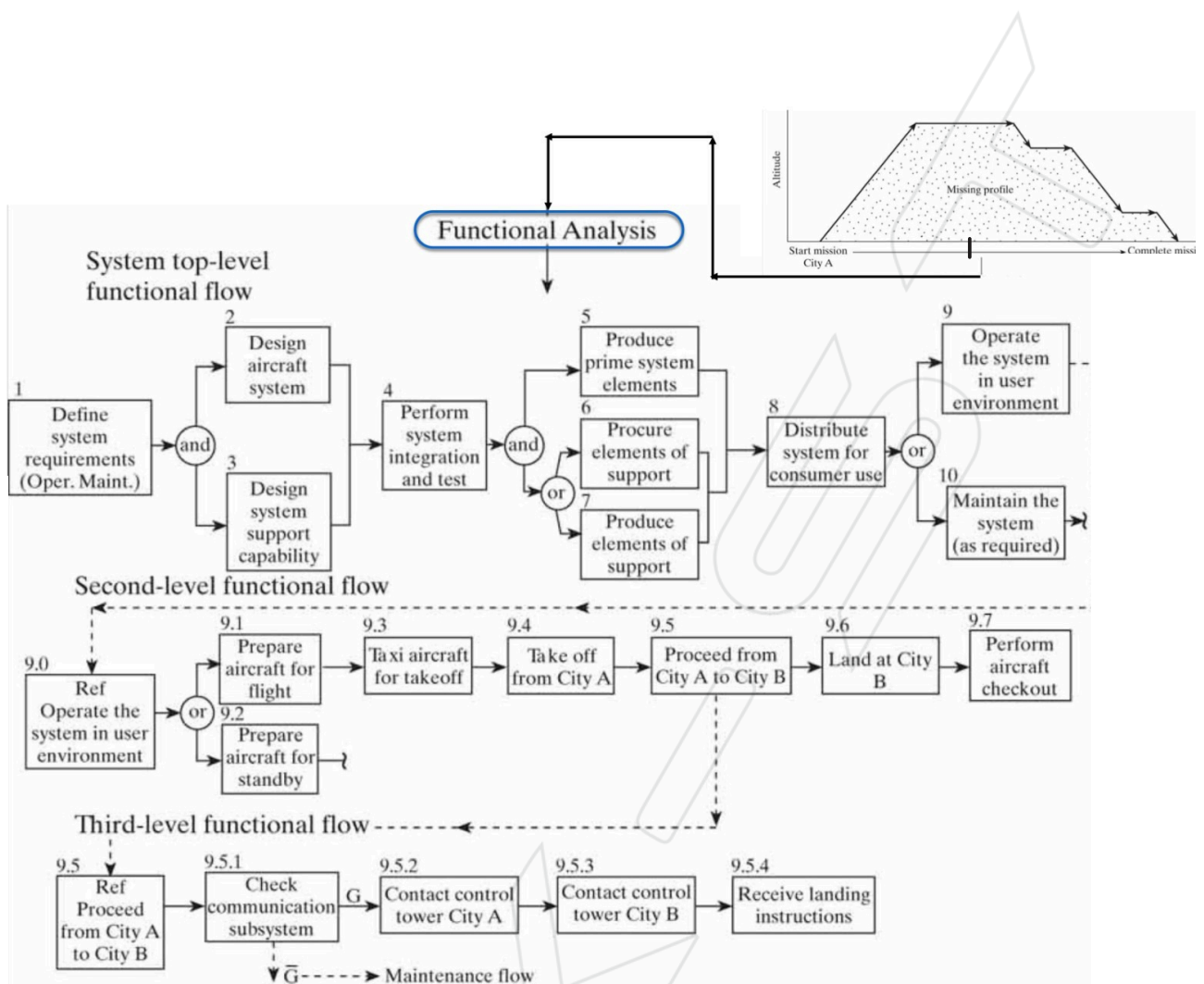
4.5 What is a CONOPS of a system? What are its main objectives?

- CONOPS means concept of operations.
- It describes 5W1H:
 - What the system will do,
 - Where it will operate,
 - When it will operate,
 - Why it is needed,
 - Who will use it,
 - How it will be used.
- Main objectives:
 - stakeholder identification and classification,
 - high-level system definition,
 - foundation of the low-level system description,
 - definition of major user classes and user activities.

Sources: Courseware 06 p.16-19; 2021.

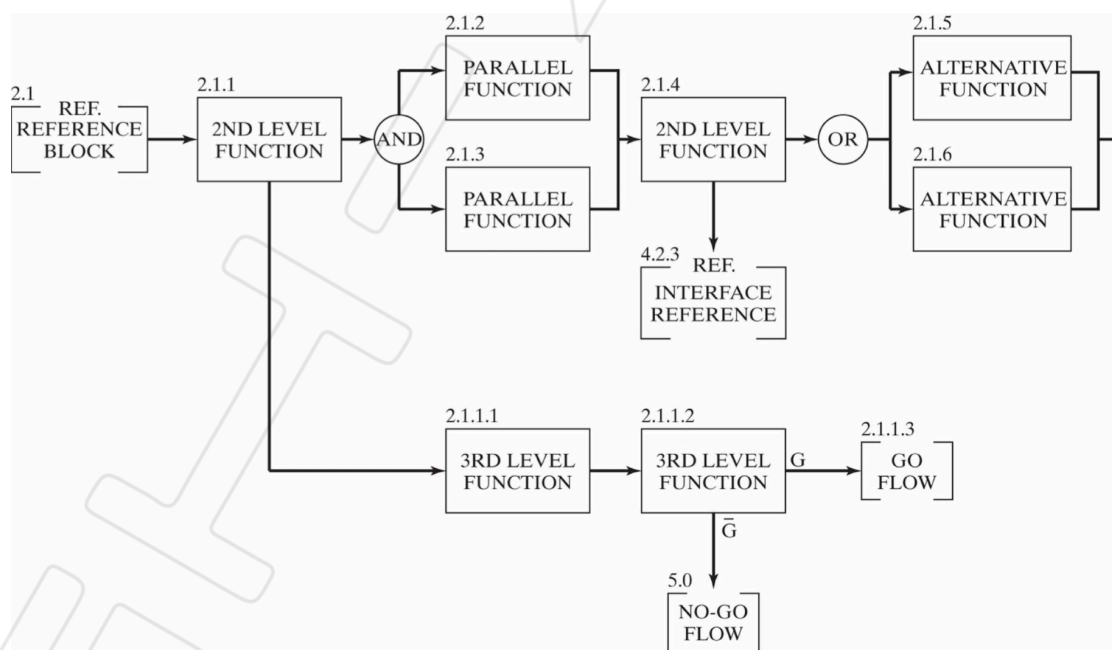
4.6 What is the purpose of functional analysis?

- Transfer requirement into coherent description of the system functions, and assign functions to system components.



Sources: Courseware 06 p.26-27, 32-36; 2020; Tutorial Solution p.1.

4.7 What is functional flow block diagram (FFBD)?



Sources: Courseware 06 p.28-36.

4.8 Technical processes are the major contributors to what stage of the lifecycle?

- Mainly to the acquisition part of the lifecycle, especially: Concept, Definition, and Design.
- Most of the work has been done by:
 - requirements analysis,
 - functional analysis,
 - architecture,
 - synthesis $\Rightarrow$ turn “what the system must do” into “what the system could be”,
 - trade-off studies,
 - early verification planning.

Sources: Courseware 06 p.2-10; Courseware 07 p.10-13.

4.9 What is technical performance measures (TPMs)?

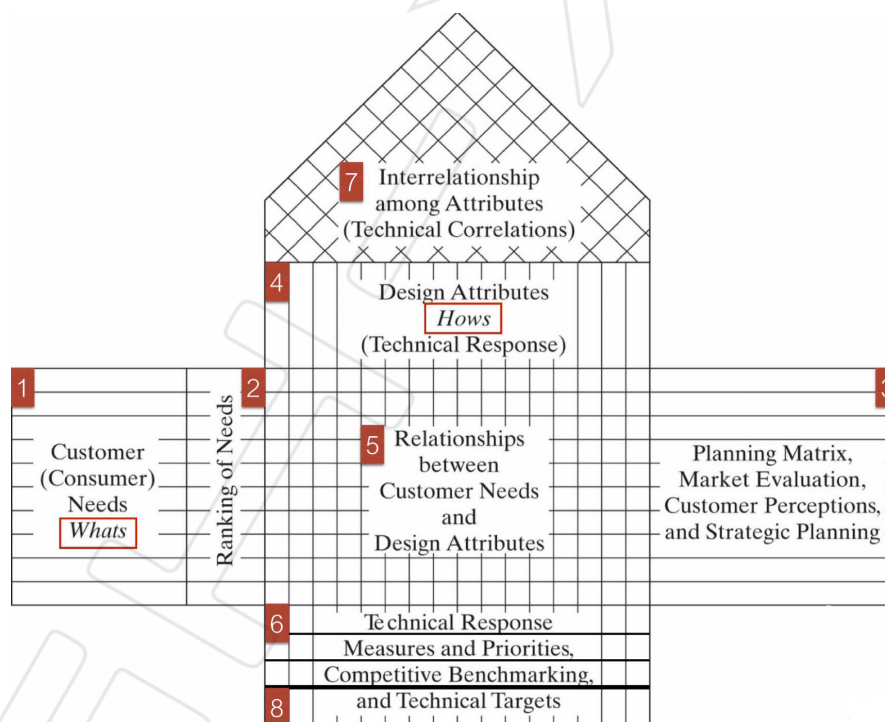
- Quantitative values that describe the system performance.

4.10 What is quality function deployment (QFD)?

- Translate customer needs into system requirements into technical solutions.

4.11 What is House of Quality (HoQ)?

- Translate customer needs into TPMs (identification and prioritisation of TPMs).



Sources: Courseware 06 p.38-41; Tutorial Solution p.7; 2020; 2022.

4.12 What is functional allocation?

- Functional allocation maps functions to hardware, software, humans, or combinations of them.

Sources: Courseware 06 p.35-36; Courseware 09 p.32, 38.

4.13 What is the purpose of system trade-off studies? At what stages of the lifecycle do you carry them out? What are the common mistakes in trade-off studies?

- Purpose:
 - synthesis $\Rightarrow$ combining components to create feasible alternatives system configurations,
 - evaluate feasible alternatives and identify the preferred design,
 - optimize a figure of merit based on stakeholder need.
- Always perform the sensitivity analysis $\Rightarrow$ which design parameters affect the system most?
- Main lifecycle stages:
 - concept,
 - definition,
 - design,
 - and later major decision points if the design changes.
- Common mistakes:
 - not stating the problem in stakeholder terms,
 - using dependent criteria,
 - vague problem statement,
 - substituting a related attribute,
 - sensitivity-analysis mistakes,
 - Forer effect (blindly adopting old criteria),
 - weight-of-importance mistakes,
 - anchoring and status quo bias,
 - treating gains and losses equally,
 - not using scoring functions,
 - implying false precision,
 - ignoring expert opinion.

Sources: Courseware 05 p.17; Courseware 06 p.47-49; Courseware 07 p.32; 2021; 2022.

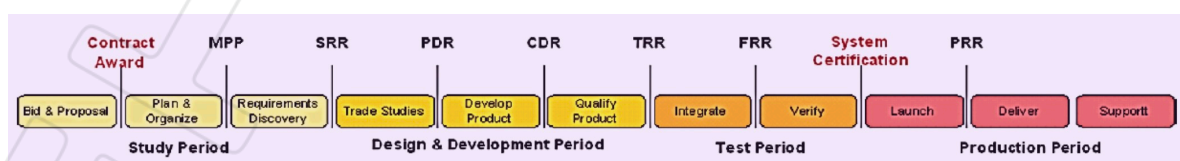
4.14 Describe the types of tests carried out during the lifecycle of a system. What are the tests that can be used to assess the true capability of the system?

- Main lifecycle test categories:
 - developmental test and evaluation (DT&E) $\Rightarrow$ acquisition,
 - acceptance test and evaluation (AT&E) $\Rightarrow$ transition between acquisition and utilisation,
 - operational test and evaluation (OT&E) $\Rightarrow$ utilisation.
- System tests during the lifecycle:
 - Analytical: virtual models, analysis and simulation before a physical system exists $\Rightarrow$ Concept, Definition,
 - Type 1: bench-test, engineering, software or service test models for component or subsystem physical design $\Rightarrow$ Definition, Design,
 - Type 2: prototype equipment test, e.g. flight simulator $\Rightarrow$ Design, Build,
 - Type 3: production system operated and evaluated as an integrated system, e.g. flight test $\Rightarrow$ Build, Test,
 - Type 4: system designed and operating in the customer environment, where true capability is first assessed $\Rightarrow$ Operate.
- Tests that best assess true system capability: OT&E
 - demonstrations with users,
 - realistic-condition flight and system tests,
 - operational data collection and analysis.

Sources: Courseware 07 p.21-24, 27-30; 2021; 2025.

4.15 What is the purpose of a review? What are the principal reviews during the lifecycle of a system?

- To verify and document that the requirements are met at the end of each stage of the lifecycle before moving to the next stage.
- A (technical) review (and audits) provides both the customer and contractor with a measure of progress towards the goal of successfully introducing system into operation:
- Principal reviews:



- maser program plan (MPP),
- system requirements review (SRR),
- system design review (SDR),
- preliminary design review (PDR),
- critical design review (CDR),
- test readiness review (TRR),
- final readiness review (FRR),
- system certification,
- production readiness review (PRR),
- functional configuration audit (FCA),
- physical configuration audit (PCA).

Sources: Courseware 05 p.16, 23; Courseware 07 p.17-19; 2021; 2025.

4.16 What is the effect of emerging properties in systems integration?

- Emergent properties are system-level effects created by interaction between components that may not have been purposely designed.
- Effects on integration:
 - unexpected behavior may appear,
 - performance may change,
 - safety may be affected,
 - suitability may change even if no subsystem is faulty by itself.

Sources: Courseware 07 p.5-6.

4.17 What types of tests are directed at verification and validation?

- Tests mainly directed at verification:
 - inspection,
 - analysis,
 - demonstration,
 - controlled test,
 - certification.
- Tests mainly directed at validation:
 - operational test and evaluation,

- trials,
- demonstrations with users or operators,
- realistic-condition flight and system tests,
- operational data collection and analysis.

Sources: Courseware 07 p.21-24, 29-30; Tutorial Solution p.9; 2025; 2024.

4.18 Describe the steps in the path to full product acceptance.

- plan testing and readiness,
- carry out DT&E through component, subsystem, and system levels,
- review test readiness and final readiness,
- carry out AT&E for formal customer acceptance,
- carry out OT&E in realistic operational use,
- complete analysis, documentation, and certification as needed,
- move into service and continuing operational support.

Sources: Courseware 07 p.10-13, 19, 28-30.

4.19 What should test planning include for verification and validation?

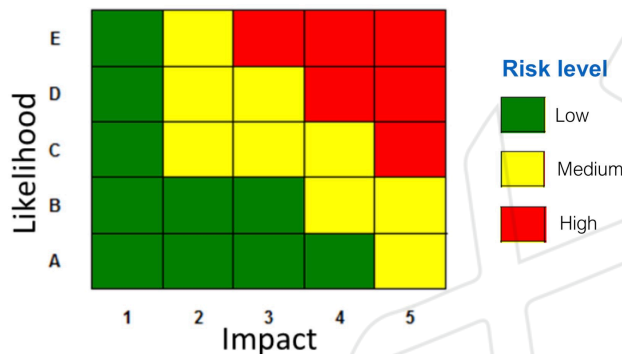
- Developmental test and evaluation (DT & E) planning should:
 - define the technical parameters to be measured,
 - summarise test events, test scenarios and the test design concept,
 - list the models and simulations to be used,
 - describe how the system environment will be represented.
- Operational test and evaluation (OT & E) planning should:
 - list the critical operational issues to determine operational suitability,
 - define technical parameters linked to those operational issues,
 - define operational scenarios and test events,
 - define the operational environment and the impact of test limitations,
 - identify test articles, logistic support and personnel training needs.
- These plans are documented in the Test and Evaluation Master Plan (TEMP).

Sources: Courseware 07 p.77; 2021.

Courseware 08 - Engineering Specialties

5.1 What is reliability, availability, hazard, risk and safety?

- Reliability: The ability of an item or system to operate under designated operating conditions for a designated period of time or number of cycles.
 - Reliability = 1 – probability of failure $\Rightarrow R(t) = P(T > t) = 1 - F(t) = 1 - P(T \leq t)$
- Availability: The probability that an item or system will be operational at a given time.
- Hazard: A physical situation with a potential for human injury, damage to property, damage to the environment, or combination of these.
- Risk: The frequency at which an individual, or a population, may be expected to sustain a given level of harm from the realisation of specified hazards.
 - Risk = likelihood $\times$ impact.
 - Risk map



- Risk analysis: risk assessment, risk management, risk communication.
- Potential source: technical, schedule, cost.
- Risk management: “avoid it”, “solve it”, or “live with it”.
- Safety: Freedom from unacceptable risk of harm.

Sources: Courseware 08 p.1-20, 45.

5.2 What is the reliability of the component after x failures expected to operate for a specified time period t? Assume an exponential density function.

- $R(x, t) = \frac{(\lambda t)^x}{x!} e^{-\lambda t}$.

Sources: Courseware 08 p.24-29, 45.

5.3 Find the reliability from components to systems.

- For components in series: $R_{\text{series}} = \prod_i^n R_i = R_a R_b \cdots R_n$
- For components in parallel: $R_{\text{parallel}} = 1 - \prod_i^n (1 - R_i) = 1 - (1 - R_a)(1 - R_b) \cdots (1 - R_n)$

Sources: Courseware 08 p.28-29.

5.4 Why is standby redundancy more reliable than concurrent redundancy? Justify it with a calculation using a system with two components in parallel.

- With two identical units and a 100% reliable switch:
 - standby redundancy: $R_{\text{standby}} = e^{-\lambda t}(1 + \lambda t)$
 - concurrent redundancy: $R_{\parallel} = 1 - (1 - e^{-\lambda t})^2$
- Standby is more reliable.

Sources: Courseware 08 p.30-35.

5.5 Think of other systems that are safe but unreliable and reliable but unsafe.

- Safe but unreliable examples:
 - a railway signal system that defaults to red for small faults and causes many unnecessary stops,
 - an emergency shutdown system that trips too often but keeps the plant safe,
 - a smoke alarm that gives frequent false alarms.
- Reliable but unsafe examples:
 - a machine that runs very consistently but has poor guarding,
 - a control system that always follows an unsafe command correctly,
 - a vehicle subsystem that almost never fails but has hazardous failure consequence when it does fail.
- Reliability and safety are related but not the same thing.

Sources: Courseware 08 p.6-7; Tutorial p.3.

Courseware 09 - Human Factors

6.1 Describe the types of human errors according to Reason. Is blaming accidents on human errors helpful to avoid them in the future? Why?

- Reason's main error types:
 - slip: unintended action caused by attentional failure,
 - lapse: unintended action caused by memory failure,
 - mistake: intended action based on wrong rule, wrong knowledge, or wrong judgement.
- Blaming accidents simply on "human error" is not very helpful because:
 - it tells you where the error happened,
 - but not why the conditions for that error existed.

Sources: Courseware 09 p.44, 48; 2025; 2020.

6.2 What is anthropometry? What is ergonomics?

- Anthropometry is the study of human body size, shape, posture...relevant to design.
- Ergonomics eliminate the design that could cause injury to people who operate, maintain or use the system.

Sources: Courseware 09 p.4, 48.

6.3 Name a few factors affecting human performance.

- Workplace factors: unsuitable workplace, housekeeping, tools, system not resilient to failure...
- Personnel factors: lack of training or experience, health, fatigue, stress, lack of motivation...
- Task factors: poorly designed, too high or low workload or time pressure...
- Organisational factors: poor leadership, poor safety culture...

Sources: Courseware 09 p.13-16, 24, 48; 2024.

6.4 How to develop the human factor requirements? At what stages of the lifecycle of the system are they required?

- Qualitative and quantitative human-factor-related requirements are developed through:
 - feasibility analysis,
 - development of operational requirements,
 - development of a maintenance concept,
 - identification and prioritization of TPMs.
- They are required across the full lifecycle.

Sources: Courseware 09 p.37-39, 47-48; Courseware 05 p.17-18.

6.5 What is the difference between an error and a violation?

- An error is unintentional or intentional but mistaken action.
- A violation is intentional action.

Sources: Courseware 09 p.44, 47-48.

6.6 What type of tasks are humans best suited for? Justify this based on human mental characteristics.

- Humans are best suited to tasks needing:
 - judgement,
 - adaptation,
 - learning,
 - broad interpretation,
 - response to novelty.

Sources: Courseware 09 p.30-32.

6.7 What is the SHEL model of human factors?

- SHEL is a human-centered model of interfaces in human factors:
 - S = Software,
 - H = Hardware,
 - E = Environment,
 - L = Liveware.

Sources: Courseware 09 p.48; 2023; 2020.

6.8 What is the most important outcome of human system integration (HSI)?

- Maximize system usability and safety.

Sources: Courseware 09 p.21, 24; 2024.

6.09 What are the questions HSI aims to answer?

- can this person,
- with this training,
- perform these tasks,
- to these standards,
- under these conditions?

Sources: Courseware 09 p.21; 2023; 2025.

Courseware 10 - Flapping-Wing

7.1 What properties of flapping-wing flight could be useful for MAVs?

- Flapping wings become especially attractive where very small scale and agile motion matter.

Sources: Courseware 10 p.4, 13, 66; 2020; 2021.

7.2 What are the most typical wing configurations used on FWMAVs, and how do they differ?

- Two wings: bird-inspired
 - less complex physics,
 - easier to design and manufacture,
 - wing motion cause large internal oscillations,
 - less wing area $\Rightarrow$ higher wing loading $\Rightarrow$ less agile.
- Tandem (four wings): insect-inspired
 - very agile,
 - complex mechanics and control.
- X-wing (four wings): common in academia
 - opposite-phase flapping $\Rightarrow$ reduce body oscillation $\Rightarrow$ more stable,
 - more lift introduce by the clap-and-fling (wing-wing interaction),
 - less lift since smaller flap altitude.

Sources: Courseware 10 p.14-18; 2020.

7.3 How do flapping-wing vehicles generate flight forces?

- By moving the wings through repeated upstroke and downstroke motion.

Sources: Courseware 10 p.19-25; 2020.

7.4 What is wing rotation and why is it important for flapping-wing MAVs?

- Wing rotation is the rapid reorientation of the wing around a spanwise axis at stroke reversal, between upstroke and downstroke.
- It is important because:
 - it keeps leading edge always meet the inflow first through the flap cycle,
 - it helps maintain positive angle of attack,
 - it generates extra circulation,
 - it can significantly increase lift (with correct timed rotation).

Sources: Courseware 10 p.19, 25.

7.5 What aerodynamic force-generation processes are involved in flapping flight?

- freestream circulation,
- rotational circulation,
- leading edge vortex and delayed stall,
- wing-wake interaction,
- wing-wing interaction (clap-and-fling),
- added mass effect.

Sources: Courseware 10 p.23-29, 35; 2023; 2020.

7.6 What is the reduced frequency?

- Reduced frequency k is a non-dimensional measure of unsteadiness in flapping-wing flight.
- $k = \frac{\omega \bar{c}}{2U_{ref}} = \frac{2\phi f \bar{c}}{2U_{ref}} = \frac{\phi f \bar{c}}{U_{ref}}$
 - low $k \Rightarrow$ more quasi-steady behavior,
 - high $k \Rightarrow$ stronger unsteady aerodynamic effects.

Sources: Courseware 10 p.30.

7.7 When is quasi-steady aerodynamic modeling considered acceptable to model flapping-wing aerodynamics?

- reduced frequency is low, about $k < 0.1$,
- forward-flight effects dominate over flapping effects,
- flapping frequency is much higher than body natural frequencies, so flap-average forces dominate body dynamics,
- the flyer is relatively rigid, often two-winged with limited wing-wing interaction.

Sources: Courseware 10 p.32-35; 2020.

7.8 What assumption is quasi-steady modeling based on?

- It assumes that the instantaneous forces on a flapping wing are the same as the forces that would act on a steadily moving wing at the same instantaneous velocity and angle of attack.
- It neglected unsteady effects.

Sources: Courseware 10 p.33.

7.9 What simplifications are often made when modeling FWMAV dynamics?

- single rigid-body model instead of full multibody model,
- time-scale separation with flap-cycle averaging,
- linearization around an equilibrium flight condition.

Sources: Courseware 10 p.39-43.

7.10 Explain the principle of time-scale separation in the context of flapping-wing flight.

- When flapping dynamics are much faster than body dynamics:
 - average over one flap cycle,
 - treat the body as reacting mainly to cycle-average forces and moments,
 - convert a time-varying problem into a simpler time-invariant one.

Sources: Courseware 10 p.41.

7.11 What is the difference between free-flight testing and wind-tunnel testing?

- Wind-tunnel (tethered or force-balance) testing:
 - better controlled conditions,
 - repeatable,
 - feasible before FWMAV is able to fly,
 - ignore the oscillation,
 - can only test steady state behaviour.
- Free-flight testing:
 - realistic, unconstrained motion, can include maneuvers,
 - need vehicle that can fly,
 - hard to control
 - hard to repeat,
 - body oscillation affect onboard sesors.

Sources: Courseware 10 p.46-47.

7.12 What are the advantages and disadvantages of tailplane-based actuation and wing-based actuation?

- Tailplane-based actuation advantages:
 - passive stability in many flight conditions,
 - simpler control,
 - simpler flapping mechanism,
 - lower overall system complexity.
- Tailplane-based actuation disadvantages:
 - less maneuverable than wing-controlled designs,

- more sensitive to perturbations such as gusts,
 - $F_{\text{tail}} \propto V^2 \Rightarrow$ steering authority drops at low airspeed,
 - no direct roll control $\Rightarrow$ yaw coupling.
- Wing-based actuation advantages:
 - higher maneuverability,
 - more direct control authority,
 - useful at low speed and in aggressive maneuvers.
 - Wing-based actuation disadvantages:
 - more complex mechanism,
 - more complex control,
 - less passive stability.

Sources: Courseware 10 p.53-54.